USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LAB #7 (Lesson 23) - Array Factor and Pattern Lab
Lab Sheet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

This sheet is the turn-in document for both parts of the Lesson 23 lab. In Part A, fill the calculated columns before you sweep and the measured columns at the bench, so the prediction is on paper before the trace is on the screen. In Part B, report every lobe amplitude relative to the main-lobe peak, in dBc, and state the uncertainty next to any number read off a hand-walked trace. The lab is a team assignment, so one sheet per team.

Part A – Learning Objective 3.2: I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

1. **The measurement table.** One row per aperture, at 10.3 GHz with $d = 14$ mm. Report each trace's peak level, and the step from the eight-element peak beside it, and report each sidelobe level relative to that trace's own peak.

Active elements	Peak (dBFS)	HPBW meas	HPBW calc	FNBW meas	FNBW calc	First SLL
8 (Rx1-Rx8)						
4 (Rx3-Rx6)						
2 (Rx4, Rx5)						

2. **Written answer 1.** Explain why the first-null beamwidth of the two-element array is quoted as 180° rather than measured. Name the condition on $\lambda/2d$ that decides whether a null exists in visible space, and evaluate it for $d = 14$ mm at 10.3 GHz.

3. **Written answer 2.** The gain of each patch element is the same whether the other seven elements are on or off, and the amplifier behind each element is unchanged. Explain why the peak of the sweep nonetheless drops by about 6 dB every time you halve the number of active elements.

Part B – Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

4. **The annotated trace.** Attach or sketch one screenshot of your best broadside run from the Tracking tab, with the main lobe and both first sidelobes labeled with their dBc values.

5. **The amplitude comparison table.** Three lobes from the broadside run, each as an amplitude relative to the main-lobe peak. The mechanical column is your hand-walked trace, the electrical column is the Rectangular-tab sweep of step B4, and the calculated column comes from theory.

Lobe	**Mechanical (dBc)**	**Electrical sweep (dBc)**	**Calculated (dBc)**	**Mechanical − electrical**
Main lobe				
First sidelobe, left				
First sidelobe, right				

6. **Written answers.** Three or four sentences each.

   (a) Why do the lobe amplitudes survive a sloppy rotation while the lobe angles do not? Refer to what the time axis does and does not record.

   (b) Which of the four artifacts of the hand measurement does an anechoic chamber remove, and which of them would still be present in a chamber measurement?

Documentation: _____